

Jharkhand University of Technology, Ranchi

Diploma 4th Semester Examination, 2026 (NEP)

Subject : SEPP

Subject Code : CSE 404

Time Allowed : 3 Hours

Full Marks : 70

Answer in your own words.

Answer any five questions in which Question No. 1 is compulsory.

The figures in the margin indicate full marks.

All questions carry equal marks.

1. Choose the correct answer of the following questions:

2×7=14

- (i) RAD stands for
 - (a) Relative Application Development
 - (b) Rapid Application Development
 - (c) Rapid Application Document
 - (d) None of the mentioned
- (ii) Which one of the following models is not suitable for accommodating any change?
 - (a) Build & Fix Model
 - (b) Prototyping Model
 - (c) RAD Model
 - (d) Waterfall Model
- (iii) Define agile scrum methodology—
 - (a) Project management that emphasizes incremental progress
 - (b) Project management that emphasizes decremental progress
 - (c) Project management that emphasizes neutral progress
 - (d) Project management that emphasizes no progress
- (iv) SDLC stands for
 - (a) Software Development Life Cycle
 - (b) System Development Life cycle
 - (c) Software Design Life Cycle
 - (d) System Design Life Cycle
- (v) Waterfall model is also called as
 - (a) Agile process model
 - (b) Classic life cycle model
 - (c) Light-Weight Model
 - (d) None of these
- (iv) Software Requirements Specification (SRS) document might include:
 - (a) Use cases
 - (b) Class diagram
 - (c) Sequence diagram
 - (d) All of these
- (vii) Who proposed the Spiral model?
 - (a) Barry Boehm
 - (b) Pressman
 - (c) Royce
 - (d) IBM

2. ~~(a)~~ What are the primary goal of software engineering? Discuss your answer with proper example.
(b) Discuss different phases of SDLC with details. 7+7
3. (a) Define the Spiral Model. What are its advantages and disadvantages? Also compare to the Agile model.
(b) Explain the various types of software testing (unit testing, integration testing, system testing, acceptance testing) with suitable examples. 7+7
4. (a) How Agile methodology is different from traditional SDLC models? Justify your answer.
~~(b)~~ Discuss Software Design. Also differentiate between high-level design and low-level design with examples. 7+7
5. ~~(a)~~ What do you mean by Data Flow Diagram (DFD)? Explain the rules of DFD and draw a Level 0 and Level 1 DFD for a college management system.
~~(b)~~ What is Requirement Analysis? Explain the different types of requirements with examples. 7+7
6. ~~(a)~~ Discuss software quality attributes. How software reliability, maintainability, and usability can play important role during development?
~~(b)~~ What is UML? Explain any five types of UML diagrams with suitable examples. 7+7
7. Write *any four* short notes: 3.5×4=14
- (a) UML
 - (b) DevOps Model
 - ~~(c)~~ Feasibility Study
 - (d) Black Box Testing
 - ~~(e)~~ Verification and Validation
 - (f) Software Reliability
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